



Lab 09 – Worksheet

Name:	ID:	Section:
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Note: Assumptions and logics should be explained separately in tasks after the task results.

Task 1

Truth Table

- a) *Main Control Unit*
- b) *ALU Control Unit*

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Comments

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Task 2

*Design Module Codes for **Main Control Unit**, and **ALU Control Unit**.*

Testbench:



Waveforms:

Task 3

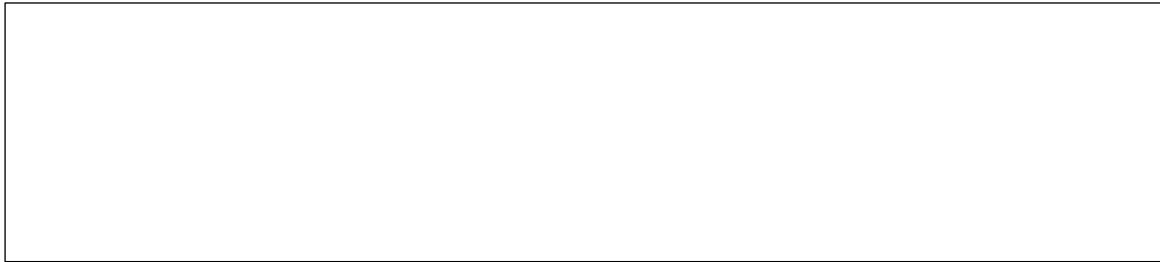
*Design module codes of **top module**:*

FSM:

Constraints File:



RTL Schematic:





Lab 9: Design and FPGA Implementation of the Control Path for a Single-Cycle RISC-V Processor

Points Distribution

Task No.	LR2 Code	LR 5 Results	LR11 Design	AR7
Task 1	-	-	/20	/20
Task 2	/20	/15		
Task 3	/10	/15		
Total Points	/80			/20
CLO Mapped	CLO 1			

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
AR9	Report	All the in-lab tasks	Most of the tasks are	Good summary of most	Detailed summary of the



		are not included in report and / or the report is submitted too late.	included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.
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